UCS420 – Cognitive Computing

Assignment 4: NumPy Introduction

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Q1. Basic NumPy Operations

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])

print("Add 2:", arr + 2)
print("Multiply by 3:", arr * 3)
print("Divide by 2:", arr / 2)
```

Q2(a). Reverse the Array

```
import numpy as np

arr = np.array([1, 2, 3, 6, 4, 5])

print("Reverse:", arr[::-1])
```

Q2(b). Most Frequent Value and Its Indices

```
import numpy as np

x = np.array([1, 2, 3, 4, 5, 1, 2, 1, 1, 1])
y = np.array([1, 1, 1, 2, 3, 4, 2, 4, 3, 3])

value_x = np.bincount(x).argmax()
index_x = np.where(x == value_x)

value_y = np.bincount(y).argmax()
index_y = np.where(y == value_y)

print("x most frequent:", value_x)
print("x indices:", index_x)
print("y most frequent:", value_y)
print("y indices:", index_y)
```

Q3. Access Elements of a 2-D Array

```
import numpy as np

arr = np.array([
    [10, 20, 30],
    [40, 50, 60],
    [70, 80, 90]
])

print("1st row, 2nd column:", arr[0, 1])
print("3rd row, 1st column:", arr[2, 0])
```

Q4. linspace(), Array Properties and Transpose

```
import numpy as np

ucs420_krish_marwaha = np.linspace(10, 100, 25)

print("Array:", ucs420_krish_marwaha)
```

```
print("Dimensions:", ucs420_krish_marwaha.ndim)
print("Shape:", ucs420_krish_marwaha.shape)
print("Total elements:", ucs420_krish_marwaha.size)
print("Data type:", ucs420_krish_marwaha.dtype)
print("Total bytes:", ucs420_krish_marwaha.nbytes)

transpose = ucs420_krish_marwaha.reshape(1, 25)
print("Transpose using reshape:")
print(transpose)

print("Using T:")
print(ucs420_krish_marwaha.T)
```

Q5. 2-D Array, Statistics, Reshape and Resize

```
import numpy as np

ucs420_krish_marwaha = np.array([
    [10, 20, 30, 40],
    [50, 60, 70, 80],
    [90, 15, 20, 35]
])

print("Array:")
print(ucs420_krish_marwaha)

print("Mean:", np.mean(ucs420_krish_marwaha))
print("Median:", np.median(ucs420_krish_marwaha))
print("Maximum:", np.max(ucs420_krish_marwaha))
print("Minimum:", np.min(ucs420_krish_marwaha))
print("Unique elements:", np.unique(ucs420_krish_marwaha))

reshaped_ucs420_krish_marwaha = ucs420_krish_marwaha.reshape(4, 3)
print("Reshaped array:")
print(reshaped_ucs420_krish_marwaha)

resized_ucs420_krish_marwaha = np.resize(ucs420_krish_marwaha, (2, 3))
print("Resized array:")
print(resized_ucs420_krish_marwaha)
```

Note: The assignment asks for an array named with the student's name. The variable name used here is ucs420_krish_marwaha. The code follows the questions in the provided Assignment-4 document.